A Descriptive Analysis of the MSR 2026 AI Development Dataset: Agent Distribution, Metadata Characteristics, and Data Quality Observations

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***Abstract*—The MSR 2026 Challenge dataset provides a large-scale collection of pull requests involving AI-assisted development. This paper presents a descriptive analysis of 50,000 pull requests sampled from the full dataset of 932,791 entries. We examine distributions of agents, metadata fields, PR titles, descriptions, file path characteristics, and keyword occurrences. We also identify data quality issues including missing fields, timestamp anomalies, duplicated title patterns, and labeling inconsistencies. Our goal is not to infer AI tool behavior or evaluate agent performance, but to provide transparent descriptive baselines and highlight dataset characteristics that may influence future research. This analysis contributes to a clearer understanding of the dataset structure and supports responsible methodological use in the MSR community.**

***Index Terms*—dataset analysis, descriptive statistics, data quality, pull requests, MSR challenge, empirical software engineering**

I. Introduction

The MSR 2026 Challenge dataset offers a rich opportunity to study AI-assisted software development at scale. However, before such data can support meaningful research, it is essential to understand its structure, distributional characteristics, and potential quality issues. This paper presents a purely descriptive examination of the dataset, focusing on metadata distributions, keyword frequencies, and observable anomalies.

Importantly, this work does not attempt to infer AI agent behavior, evaluate tool performance, or compare software engineering practices. Observational pull request data does not support such inferences without controlled experimental designs or validated attribution signals. Instead, our objective is to document the dataset's properties and provide baselines that support future methodological and empirical studies.

A. Motivation

Large-scale datasets require careful characterization before analysis. Understanding dataset properties is crucial for:

- **Methodological Design**: Researchers need to understand data limitations
- **Sampling Strategies**: Distribution imbalances affect statistical analysis
- **Quality Assessment**: Data anomalies must be documented
- **Baseline Establishment**: Descriptive statistics support future work

B. Research Questions

We address four descriptive research questions:

RQ1: *What is the distribution of agents, timestamps, titles, descriptions, and file metadata in the MSR 2026 dataset?*

RQ2: *What patterns appear in PR metadata such as title length, description availability, and file path characteristics?*

RQ3: *What data quality issues or anomalies can be observed, such as missing fields, inconsistent timestamps, or duplicate patterns?*

RQ4: *What descriptive baselines can support future researchers working with the dataset?*

C. Key Contributions

This paper makes several descriptive contributions:

1) **Dataset characterization** documenting distributions and metadata patterns across 50,000 pull requests
2) **Data quality assessment** identifying anomalies, missing fields, and potential issues
3) **Descriptive baselines** providing reference statistics for future research
4) **Methodological documentation** for responsible dataset usage
5) **Transparency framework** supporting reproducible MSR research

II. Related Work

A. AI-Assisted Software Development

Recent studies have examined AI-generated code quality [1], with Chen et al. introducing HumanEval benchmarks for code generation models. Austin et al. [2] explored program synthesis capabilities, while Nijkamp et al. [3] developed CodeGen for multi-turn programming tasks.

This work analyzes the MSR 2026 Challenge dataset. Research conducted at Edinburgh Napier University, November 2025.

These studies focus primarily on functional correctness. Our work provides dataset documentation and descriptive baselines for the MSR research community.

B. Empirical Software Engineering

Traditional empirical studies have examined code review effectiveness [7] and developer productivity [4]. Recent work has utilized large-scale datasets to understand software development patterns.

Our study contributes dataset documentation and descriptive analysis for future MSR research.

C. Software Testing and Quality

Niu [8] examined software evolution patterns, while Inozemtseva and Holmes [9] studied coverage metrics. These studies highlight the need for careful dataset characterization.

III. Methodology

A. Dataset and Sampling Strategy

We analyzed the MSR 2026 Challenge dataset containing 932,791 pull requests. Due to computational constraints with the full dataset (several GB), we employed random sampling to select 50,000 PRs using pandas `sample(n=50000, random_state=42)` for reproducibility.

Agent Distribution: The sample exhibited the following distribution:

- OpenAI Codex: 43,668 PRs (87.3%)
- GitHub Copilot: 2,751 PRs (5.5%)
- Cursor: 1,747 PRs (3.5%)
- Devin: 1,571 PRs (3.1%)
- Claude Code: 263 PRs (0.5%)

This distribution reflects the characteristics present in the full dataset.

B. Keyword Detection Methodology

We implemented keyword-based detection for the presence of testing-related terms:

```
test_keywords = ['test', 'spec', 'unittest', 'pytest',
                 'jest', 'karma', 'mocha', 'cypress',
                 'testing', 'junit', 'testng', 'rspec',
                 'phpunit', 'nunit', 'xunit']

def contains_test_keyword(row):
    title = str(row.get('title', '')).lower()
    body = str(row.get('body', '')).lower()
    return any(keyword in title or keyword in body
               for keyword in test_keywords)
```

This approach detects explicit mentions of testing-related terminology in PR metadata. It indicates only keyword presence in text fields.

C. Descriptive Analysis Framework

We employed purely descriptive analysis:

Frequency Counts: Calculated keyword occurrence rates and field distributions for each agent label.

Basic Statistics: Computed means, medians, and ranges for numerical fields like title length.

Missing Data Analysis: Identified null values, empty fields, and data quality issues.

No Inferential Statistics: We use only descriptive statistics. This analysis makes no causal claims or inferential comparisons.

D. Data Quality Assessment

Descriptive analysis revealed several dataset characteristics:

- Missing PR descriptions in 1,847 cases (3.7%)
- Timestamp inconsistencies (12 PRs predating 2024)
- Agent labels of unknown accuracy or validation method
- Duplicate titles with different agent labels (23 cases)
- Varying PR title lengths and description completeness

We document these characteristics without modification to provide transparency about dataset properties. These observations highlight the importance of understanding data limitations before conducting research.

IV. Results

A. RQ1: Agent Distribution and Metadata Patterns

Table I presents descriptive statistics for the 50,000 PR sample, showing keyword occurrence frequencies and metadata characteristics by agent label.

TABLE I
DESCRIPTIVE STATISTICS FOR MSR 2026 DATASET SAMPLE

Agent	Count	Keyword %	Closed %	Users	Avg Title
OpenAI Codex	43,668	98.5	93.1	13,421	67.2 chars
GitHub Copilot	2,751	80.0	91.8	1,203	53.4 chars
Claude Code	263	80.2	94.3	124	59.1 chars
Devin	1,571	67.1	89.4	542	48.3 chars
Cursor	1,747	18.0	88.7	687	42.1 chars
Overall	**50,000**	**93.6**	**92.3**	**15,667**	**62.4 chars**

Note: "Keyword %" indicates the percentage of PRs containing testing-related keywords in titles or descriptions.

Descriptive Statistics:

- **OpenAI Codex**: 98.5% of PRs contain testing keywords (42,997/43,668 PRs)
- **GitHub Copilot**: 80.0% of PRs contain testing keywords (2,201/2,751 PRs)
- **Claude Code**: 80.2% of PRs contain testing keywords (211/263 PRs)
- **Devin**: 67.1% of PRs contain testing keywords (1,054/1,571 PRs)
- **Cursor**: 18.0% of PRs contain testing keywords (314/1,747 PRs)
- **Overall**: 93.6% of sampled PRs (46,777/50,000) contain testing keywords

These values characterize the text fields in the dataset and do not indicate development practices or tool behavior. We report frequencies without interpreting differences.

B. RQ2: Metadata Characteristics

Closure Status: PR closure rates by agent label are: OpenAI Codex (93.1%), GitHub Copilot (91.8%), Claude Code (94.3%), Devin (89.4%), and Cursor (88.7%).

User Counts: Unique user counts by agent label are: OpenAI Codex (13,421 users), GitHub Copilot (1,203 users), Claude Code (124 users), Devin (542 users), and Cursor (687 users).

Title Lengths: Average title lengths by agent label are: OpenAI Codex (67.2 characters), GitHub Copilot (53.4 characters), Claude Code (59.1 characters), Devin (48.3 characters), and Cursor (42.1 characters).

C. RQ3: Data Quality Observations

Several data quality characteristics were observed:

Missing Descriptions: 1,847 PRs (3.7%) lack description content.

Timestamp Anomalies: 12 PRs show timestamps predating 2024.

Duplicate Patterns: 23 pairs of PRs share identical titles but different agent labels.

Label Validation: Agent labels appear to be heuristically assigned without verified accuracy documentation.

These observations highlight the importance of data preprocessing and quality assessment when using the MSR 2026 dataset for research.

D. RQ4: Descriptive Baselines for Future Research

The following baseline statistics may assist future researchers:

- **Dataset Size**: 932,791 total PRs with 50,000 analyzed
- **Agent Imbalance**: 87.3% OpenAI Codex, other agents ¡6% each
- **Keyword Frequency**: 93.6% overall testing keyword presence
- **Data Completeness**: 96.3% PRs have descriptions, 99.98% have valid timestamps

E. Summary of Descriptive Findings

The 50,000 PR sample contains 15,667 unique users (3.19 PRs per user average). The dataset spans December 2024 through July 2025. Agent label distribution shows OpenAI Codex representing 87.3% of entries. These characteristics may inform sampling and analysis strategies in future research.

V. Discussion

A. Dataset Characteristics for Future Research

Our descriptive analysis reveals several important dataset properties:

Agent Label Distribution: OpenAI Codex represents 87.3% of the dataset sample. Researchers should consider this distribution when designing sampling strategies.

Keyword Frequencies: Testing-related keyword frequencies are documented in Table I for reference in future research.

Data Quality Considerations: Missing descriptions (3.7%), timestamp anomalies, and duplicate titles should be considered in preprocessing steps for future analyses.

User Counts: The dataset contains 15,667 unique users across 50,000 PRs, which may affect independence assumptions in statistical modeling.

B. Methodological Implications

This descriptive analysis contributes to responsible dataset usage:

Baseline Establishment: The descriptive statistics provide reference baselines for future MSR research using this dataset.

Sampling Guidance: The observed imbalances inform appropriate sampling strategies for balanced comparative analyses.

Quality Assessment Framework: The data quality issues we identified suggest preprocessing steps that future researchers should consider.

C. Scope and Limitations

Descriptive Scope:

- **Keyword Detection**: Our analysis detects only explicit keyword mentions, not actual testing practices or quality
- **Agent Labels**: Dataset labels are of unknown accuracy and validation—we report them as provided without verification
- **Sample Coverage**: 50K sample represents 5.4% of the full dataset

No Behavioral Inference:

- **No Causal Claims**: This analysis makes no claims about AI agent behavior, capabilities, or performance
- **No Quality Assessment**: Closed PR status and keyword frequency do not indicate code quality or testing effectiveness
- **Metadata Only**: Analysis is limited to PR metadata without examining actual code changes

Dataset Limitations:

- **Platform Scope**: Data comes from GitHub repositories only
- **Label Reliability**: Agent attribution method and accuracy are undocumented
- **Temporal Scope**: Eight-month period (December 2024 - July 2025) with potential incomplete coverage

D. Future Research Opportunities

This descriptive analysis suggests several research directions:

Validation Studies: Future work could validate agent label accuracy through manual annotation or alternative detection methods.

Code-Level Analysis: Examining actual code changes rather than metadata could provide more detailed dataset characterization.

Comparative Datasets: Comparing MSR 2026 dataset characteristics with other AI development datasets.

Preprocessing Guidelines: Developing standardized preprocessing approaches for handling missing data and quality issues.

Methodological Studies: Using this dataset to study appropriate methodological approaches for observational software engineering research.

VI. Conclusion

This descriptive analysis of 50,000 pull requests from the MSR 2026 dataset documents agent distribution, metadata characteristics, and data quality issues. Testing-related keyword frequencies are documented by agent label in Table I.

Key Documentation:

1) Agent label distribution documented (87.3% OpenAI Codex)
2) Testing keyword frequencies documented by agent label
3) Data quality issues include missing descriptions (3.7%) and timestamp anomalies
4) 15,667 unique users contributed to the 50,000 PR sample
5) Baseline statistics are established for future research reference

Research Support: This analysis provides transparent documentation of dataset characteristics, identifies data quality considerations, and establishes descriptive baselines for the MSR research community.

Methodological Contribution: By focusing on descriptive analysis rather than behavioral inference, this work demonstrates responsible approaches to observational dataset analysis and highlights the importance of understanding data limitations.

Future research can build on these baselines while addressing the methodological and data quality considerations we have identified.

Data Availability

The MSR 2026 Challenge dataset is publicly available. Our analysis scripts and processed results are available at: https://github.com/ghost1100/MSR/

Acknowledgments

We thank the MSR 2026 Challenge organizers for providing the dataset and Edinburgh Napier University for supporting this research.

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